

DURAND DYER-BRANCH

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Education

San Francisco State University

Aug. 2023 – May 2027

Bachelor of Science in Computer Science

San Francisco, CA

– **Coursework:** Data Structures, Algorithms, Internet Technology, Web Development, Software Engineering

Technical Skills

Languages: C++, Java, Python, JavaScript, HTML/CSS

Technologies: C++/WinRT, WinUI 3, Win32 API, Spring Boot, FastMCP, React, Node.js, Redis, PostgreSQL, Docker, GCP, Google ADK

Tools: CMake, Visual Studio, Vim, JetBrains IDEs, Git/GitHub, GitHub Actions, Postman

Experience

Daisy Refillery

Jan. 2024 – June 2024

Full-Stack Developer

San Francisco, CA

- Revamped the React and JavaScript storefront, reducing bounce rate by 18% and increasing product-page traffic 25% month over month.
- Built Node.js/Express APIs and PostgreSQL integrations that automated inventory updates, eliminated two hours of daily manual work, and raised data accuracy to 99%.
- Coordinated Git workflows for a six-person team and introduced GitHub Actions CI/CD, increasing deployment frequency by 50%.

Projects

Account Armory | C++, C++/WinRT, WinUI 3, Win32 API

Aug. 2026 – Present

- Built a 40-file credential manager with separate launcher and website vaults, responsive views, CRUD, search, filtering, and CSV imports for four browser formats.
- Protected sensitive actions with Windows DPAPI and Windows Hello; added configurable auto-lock and conditional clipboard clearing.
- Designed AES-256-GCM backups and versioned JSON persistence with atomic writes, rollback, crash recovery, and corrupt-data quarantine.

Pet Adoption Platform | Java, Python, Spring Boot, FastMCP, Redis, Docker

May 2025 – Dec. 2025

- Architected a Spring Boot REST API with 12+ endpoints and 20+ filters; Redis caching cut external requests by 80% and delivered sub-200 ms cached responses.
- Developed four FastMCP servers exposing 19 tools over STDIO and HTTP with asynchronous requests, validation, rate limiting, structured errors, and Pytest coverage.

C++ Behavior Playground | C++20, CMake, Clang/GCC

Feb. 2026 – Present

- Built 109 focused experiments covering object representation, conversions, pointers, references, const correctness, type deduction, arrays, strings, vectors, and iterators.
- Created CMake tooling that discovers nested topics and generates one executable per experiment, plus a harness for predictions and defined, implementation-defined, and undefined behavior classification.

Sauerbraten Win32 Memory Lab | C++20, Win32 API, DLL, Visual Studio

Aug. 2026

- Built an in-process DLL in an open-source game lab to practice module-base resolution, pointer arithmetic, memory offsets, worker-thread coordination, diagnostics, resource cleanup, and safe unloading.

Research Experience

Elliq AI Companion Evaluation

Aug. 2026 – Dec. 2026

Undergraduate Product Researcher, San Francisco State University

San Francisco, CA

- Conduct a faculty-supervised evaluation of Elliq across 25+ usability, accessibility, reliability, information-retrieval, and safety criteria.
- Test voice interaction, reminders, caregiver communication, information retrieval, and responses to simulated distress; record strengths and limitations through structured observations.
- Assess privacy, emotional dependency, and support for real-world human contact; synthesize structured findings for faculty review and potential publication.